

§4.4

Conditional Probability and the Multiplication Rule

How new information changes a probability

Today's Goal

- Understand and calculate conditional probabilities
- Apply the multiplication rule for finding $P(A \text{ and } B)$
- Use tree diagrams to solve multi-stage probability problems
- Distinguish between “with replacement” and “without replacement” scenarios

What You Will Be Graded On

LT.7 Probability

I understand what the probability of an event represents and can apply basic rules of probability to calculate the probability of an event.

Drawing One Card

A card is drawn from a standard deck of 52 cards. Before you see it, a friend tells you the card is red.

What is the probability that the card is a heart now?

Conditional Probability

Definition 4.4.1

The *conditional probability* of event B given that event A has occurred, denoted $P(B \mid A)$ (read “the probability of B given A ”), is the probability that B occurs assuming that A has already occurred. It is computed by

$$P(B \mid A) = \frac{P(A \text{ and } B)}{P(A)}, \quad P(A) > 0.$$

Probability of a Heart, Given Red

Before the hint, all 52 cards are possible, and 13 of them are hearts:

$$P(\text{heart}) = \frac{13}{52} = \frac{1}{4}.$$

After the hint, only the 26 red cards are possible. All 13 hearts are red:

$$P(\text{heart} \mid \text{red}) = \frac{13}{26} = \frac{1}{2}.$$

With the formula: $P(\text{heart} \mid \text{red}) = \frac{13/52}{26/52} = \frac{13}{26} = \frac{1}{2}.$

Conditional Probability From Counts

In a club of 50 members, 30 are juniors and 20 are seniors. Of the 30 juniors, 18 have a part-time job. If we randomly select a member who is a junior, what is the probability that the member has a part-time job?

Let A = member is a junior and B = member has a part-time job. We want $P(B \mid A)$, the probability of having a part-time job *given* the member is a junior.

Since we know the member is a junior, we restrict our attention to the 30 juniors. Of these, 18 have a part-time job:

$$P(B \mid A) = \frac{18}{30} = 0.60.$$

Conditional Probability by the Formula

Alternatively, using the formula:

- $P(A) = \frac{30}{50}$.
- $P(A \text{ and } B) = \frac{18}{50}$ (junior AND has a part-time job).

$$P(B \mid A) = \frac{P(A \text{ and } B)}{P(A)} = \frac{18/50}{30/50} = \frac{18}{30} = 0.60.$$

Conditional Probability From a Table

The table below shows the results of a survey of 400 shoppers.

	Spent over \$50	Spent \$50 or less	Total
Used a coupon	56	24	80
No coupon	104	216	320
Total	160	240	400

(a) $P(\text{Spent over \$50} \mid \text{Used a coupon})$.

Of the 80 coupon users, 56 spent over \$50: $\frac{56}{80} = 0.70$.

(b) $P(\text{Used a coupon} \mid \text{Spent \$50 or less})$.

Of the 240 who spent \$50 or less, 24 used a coupon: $\frac{24}{240} = 0.10$.

Conditional Probability From a Table

	Spent over \$50	Spent \$50 or less	Total
Used a coupon	56	24	80
No coupon	104	216	320
Total	160	240	400

In part (a), $P(\text{Spent over \$50} \mid \text{Used a coupon}) = 0.70$. Is $P(\text{Used a coupon} \mid \text{Spent over \$50})$ **also 0.70?**

Of the 160 who spent over \$50, 56 used a coupon: $\frac{56}{160} = 0.35$.

Each time, we restrict our attention to the group we are given: the 80 coupon users in part (a), and the 160 who spent over \$50 here.

The Multiplication Rule

The multiplication rule allows us to find the probability that two events both occur.

Theorem 4.4.4 General Multiplication Rule

For any two events A and B :

$$P(A \text{ and } B) = P(A) \cdot P(B \mid A).$$

To find the probability that both A and B occur, we multiply the probability that A occurs by the probability that B occurs *given* that A has occurred.

Two Pens Without Replacement

A cup contains 7 black pens and 5 blue pens. Two pens are drawn without replacement. Find the probability that both pens are black.

Let A = first pen is black and B = second pen is black.

- $P(A) = \frac{7}{12}$ (7 black out of 12 total).
- $P(B \mid A) = \frac{6}{11}$ (after removing a black pen, 6 black remain out of 11 total).

$$P(A \text{ and } B) = P(A) \cdot P(B \mid A) = \frac{7}{12} \cdot \frac{6}{11} = \frac{42}{132} = \frac{7}{22} \approx 0.318.$$

A Face Card, Then an Ace

Two cards are drawn from a standard deck without replacement. Find the probability that the first card is a face card and the second card is an ace.

Let A = first card is a face card and B = second card is an ace.

- $P(A) = \frac{12}{52}$ (12 face cards in a deck of 52).
- $P(B \mid A) = \frac{4}{51}$ (4 aces remain in 51 cards).

$$P(A \text{ and } B) = \frac{12}{52} \cdot \frac{4}{51} = \frac{48}{2652} = \frac{4}{221} \approx 0.018.$$

With vs. Without Replacement

The distinction between sampling *with replacement* and *without replacement* is crucial in probability calculations.

With Replacement

- After each selection, the item is returned before the next selection.
- The sample space remains the same for each selection.
- Events are independent.

Without Replacement

- Items are not returned after selection.
- The sample space changes after each selection.
- Events are dependent.

Pens With and Without Replacement

In Example 3, two pens were drawn without replacement, and $P(\text{both black}) = \frac{7}{22} \approx 0.318$.

What if the first pen is returned before the second pen is drawn?

After drawing the first pen, it is returned, so the probabilities remain the same:

$$\frac{7}{12} \cdot \frac{7}{12} = \frac{49}{144} \approx 0.340.$$

With replacement: 0.340. Without replacement: 0.318.

Getting to Work on Time

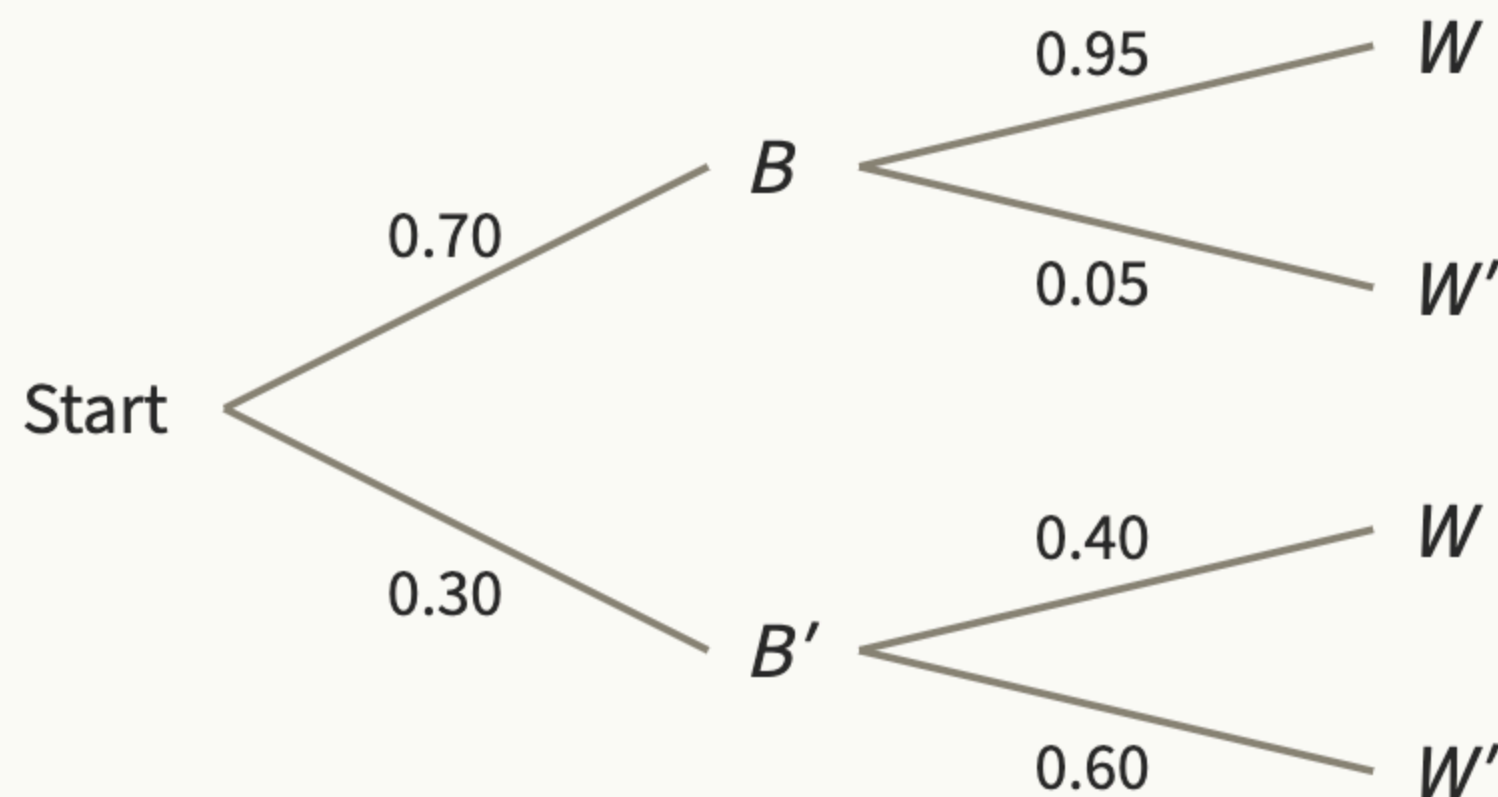
A commuter's bus is on time 70% of the time. If the bus is on time, there is a 95% chance the commuter gets to work on time. If the bus is late, there is only a 40% chance.

Is the chance of getting to work on time closer to 95% or to 40%?

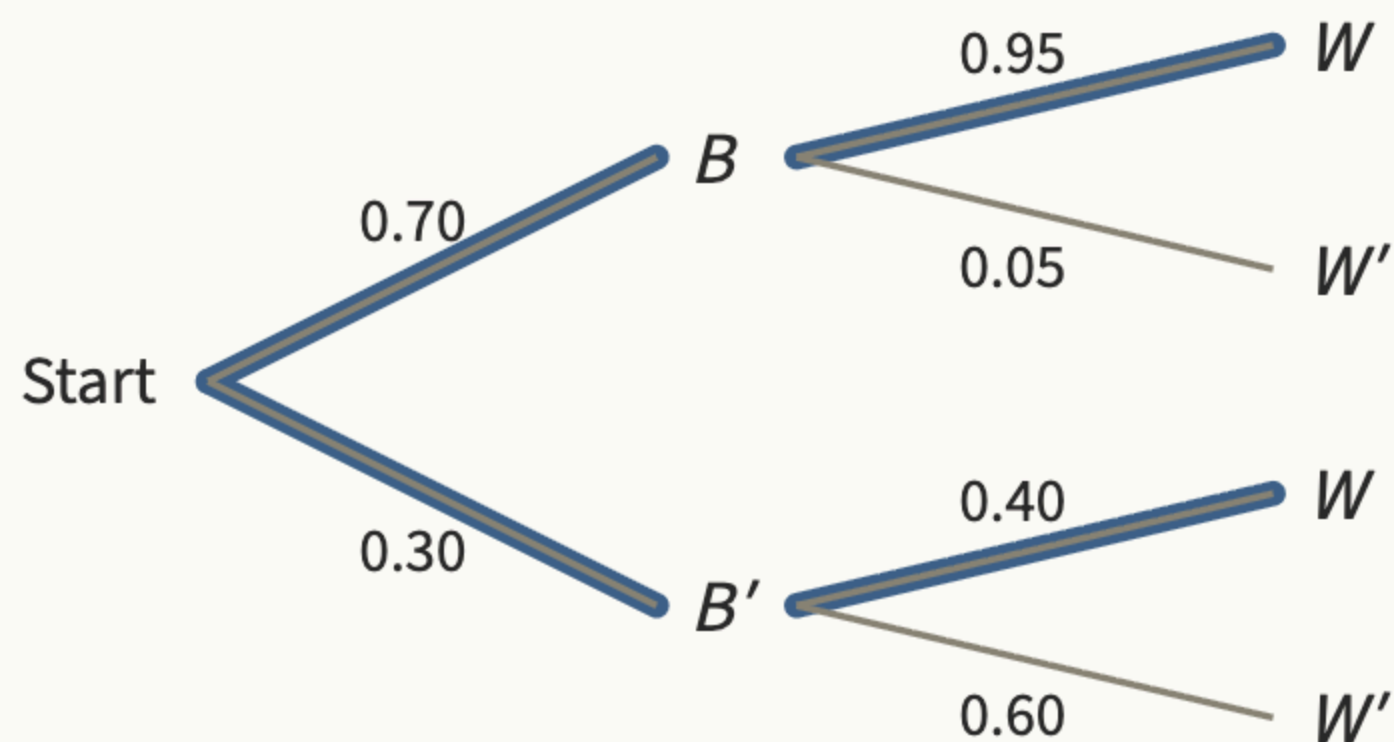
Building a Tree Diagram

A commuter's bus is on time 70% of the time. If the bus is on time, there is a 95% chance the commuter gets to work on time. If the bus is late, there is only a 40% chance. What is the probability that the commuter gets to work on time?

Let B = the bus is on time and W = gets to work on time.



Using the Tree Diagram



Two paths lead to getting to work on time.

Path 1. Bus on time, then to work on time:

$$P(B \text{ and } W) = 0.70 \times 0.95 = 0.665$$

Path 2. Bus late, then to work on time:

$$P(B' \text{ and } W) = 0.30 \times 0.40 = 0.12$$

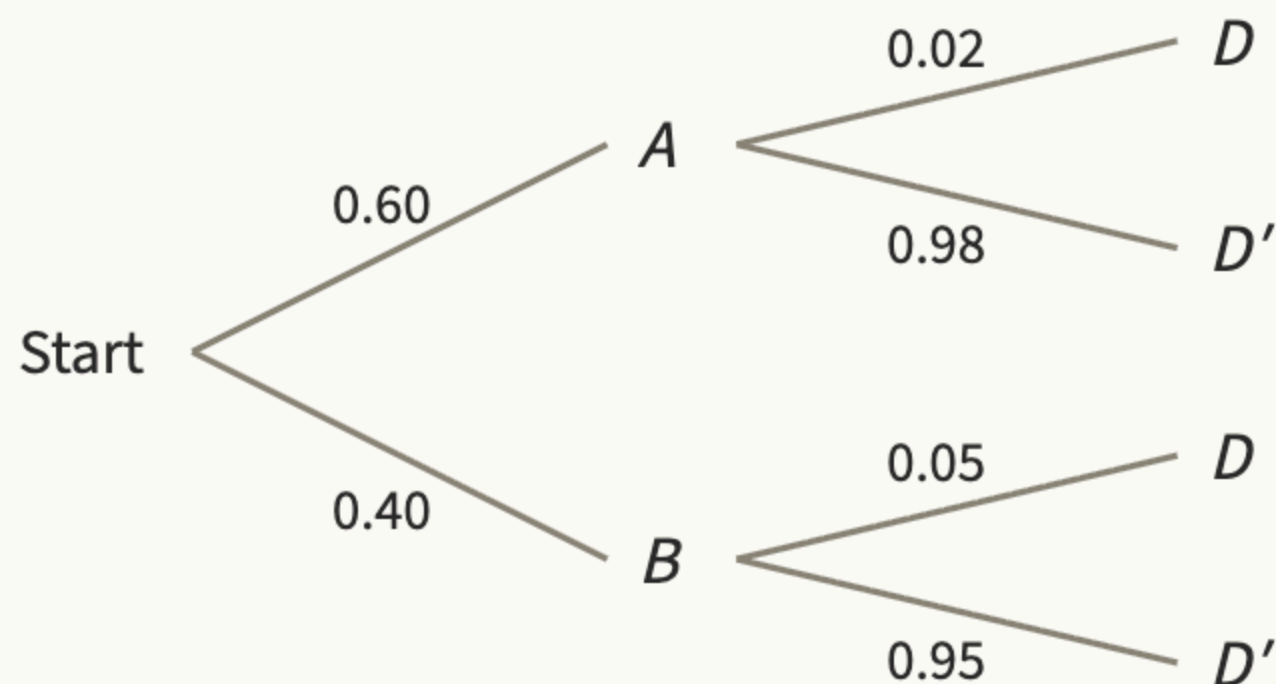
$$P(W) = 0.665 + 0.12 = 0.785$$

There is a 78.5% chance the commuter gets to work on time.

Now You Try: Two Factories

You Try 4.4.3. A company has two factories. Factory A produces 60% of the products and Factory B produces 40%. The defect rate at Factory A is 2% and the defect rate at Factory B is 5%. What is the probability that a randomly selected product is defective?

You Try 4.4.3: Checking Your Answer



$$P(A \text{ and } D) = 0.60 \times 0.02 = 0.012$$

$$P(B \text{ and } D) = 0.40 \times 0.05 = 0.020$$

$$P(\text{defective}) = 0.012 + 0.020 = 0.032$$

That is a 3.2% chance.

Extending the Multiplication Rule

The multiplication rule extends to three or more events. For three events:

$$P(A \text{ and } B \text{ and } C) = P(A) \cdot P(B \mid A) \cdot P(C \mid A \text{ and } B).$$

Three cards are drawn from a standard deck without replacement. Find the probability that all three cards are black.

Three Black Cards

Three cards are drawn from a standard deck without replacement. Find the probability that all three cards are black.

Let A = first card is black, B = second card is black, and C = third card is black.

- $P(A) = \frac{26}{52}$ (26 black cards in 52 cards).
- $P(B \mid A) = \frac{25}{51}$ (25 black cards left in 51 cards).
- $P(C \mid A \text{ and } B) = \frac{24}{50}$ (24 black cards left in 50 cards).

$$P(A \text{ and } B \text{ and } C) = \frac{26}{52} \cdot \frac{25}{51} \cdot \frac{24}{50} = \frac{15600}{132600} = \frac{2}{17} \approx 0.1176.$$

Exactly One Burned-Out Bulb

A box contains 3 burned-out bulbs and 7 working bulbs. Two bulbs are drawn without replacement. Find the probability that exactly one of the two bulbs is burned out.

In what order can exactly one burned-out bulb be drawn?

- Burned out, then working: $\frac{3}{10} \cdot \frac{7}{9} = \frac{21}{90}$.
- Working, then burned out: $\frac{7}{10} \cdot \frac{3}{9} = \frac{21}{90}$.

The two orders cannot both happen, so add: $\frac{21}{90} + \frac{21}{90} = \frac{42}{90} = \frac{7}{15} \approx 0.467$.

What to Take With You

- In $P(B \mid A)$, the group that is given becomes the new denominator.
- $P(B \mid A)$ and $P(A \mid B)$ are usually different numbers.
- $P(A \text{ and } B) = P(A) \cdot P(B \mid A)$. Without replacement, the second draw changes.
- On a tree diagram, multiply along one path. Add across different paths.